

EP3120 - Statistical Mechanics

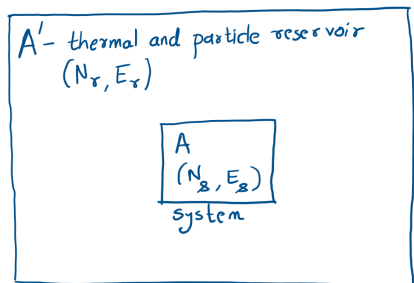
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Grand Canonical Ensembles

Systems in thermal and particle bath

- Consider a small system A in contact with a heat and particle reservoir A' , with the system much smaller than the reservoir
- The reservoir is at a fixed temperature that is the same as temperature of the system
- The reservoir can exchange energy and particles with the system
- There are many examples of such a system
- An open glass of water in a room can have evaporation/condensation of water molecules
- Radiation in a black-body can be thought of as made out of photons which can be absorbed/emitted by the blackbody
- Chemical reactions: ex. $N_2 + 3H_2 \rightleftharpoons 2NH_3$ in which case the system is a collection of ammonia molecules diffused throughout a bath of nitrogen and hydrogen

System in thermal and particle bath



- Consider the system A with N_s particles and energy E_s in a reservoir A' with N_r particles and energy E_r
- The total energy and number of particles is fixed

$$N_s + N_r = N_0 \quad (1)$$

$$E_s + E_r = E_0 \quad (2)$$

- As for canonical ensemble, we assume

$$N_s \ll N_r; \quad E_s \ll E_r \quad (3)$$

Grand canonical ensemble

- Consider an ensemble of system+bath (microcanonical)
- The probability of finding a particle microstate of the system in this ensemble is proportional to the number of microstates available to the bath

$$P_s \propto \Omega_B(N_0 - N_s, E_0 - E_s) \quad (4)$$

- The collection of all the distinct microstates of the system forms a grand canonical ensemble and the probability of finding a particular microstate in this grand-canonical ensemble is as given above
- The Taylor series expansion gives

$$\Omega_B(N_0 - N_s, E_0 - E_s) = \Omega_B(N_0, E_0) - E_s \frac{\partial \Omega_B}{\partial E} - N_s \frac{\partial \Omega_B}{\partial N} \quad (5)$$

$$+ \frac{E_s^2}{2} \frac{\partial^2 \Omega_B}{\partial E^2} + \frac{N_s^2}{2} \frac{\partial^2 \Omega_B}{\partial N^2} + E_s N_s \frac{\partial^2 \Omega_B}{\partial E \partial N} + \dots \quad (6)$$

Grand canonical ensemble

- As in the canonical ensemble, we can use the relation

$$\frac{\partial S}{\partial E} = \frac{1}{T} \implies \frac{\partial \Omega}{\partial E} = \beta \Omega \quad (7)$$

- Using the relation $S = k_B \ln \Omega$ we have

$$\frac{\partial \Omega}{\partial N} = \frac{\Omega}{k_B} \frac{\partial S}{\partial N} \quad (8)$$

- Introduce a new quantity α such that

$$\left. \frac{\partial \Omega}{\partial N} \right|_{V,T} = \alpha \Omega \quad (9)$$

Grand canonical ensemble

- As seen earlier, the Ω will vary much more strongly with E and N compared to the variation of β and α

$$\frac{\partial^2 \Omega}{\partial N^2} \approx \alpha \frac{\partial \Omega}{\partial N} = \alpha^2 \Omega \quad (10)$$

$$\frac{\partial^2 \Omega}{\partial N \partial E} \approx \beta \frac{\partial \Omega}{\partial N} = \alpha \beta \Omega \quad (11)$$

$$\frac{\partial^2 \Omega}{\partial E^2} \approx \beta \frac{\partial \Omega}{\partial E} = \beta^2 \Omega \quad (12)$$

- Substituting in the Taylor series gives

$$\Omega(N_0 - N_s, E_0 - E_s) = \Omega(N_0, E_0) \left[1 - \beta E_s - \alpha N_s \right. \quad (13)$$

$$\left. + \frac{\beta^2 E_s^2}{2} + \frac{\alpha^2 N_s^2}{2} + \beta \alpha N_s E_s + \dots \right] \quad (14)$$

$$= \Omega_B[N_0, E_0] \exp[-\beta E_s - \alpha N_s] \quad (15)$$

Grand canonical partition function

- From this we get the probability of a microstate s in the grand canonical ensemble as

$$P_s \propto e^{-\beta E_s - \alpha N_s} \quad (16)$$

- The proportionality constant is given as

$$P_s = \frac{1}{\mathcal{Z}} e^{-\beta E_s - \alpha N_s} \quad (17)$$

- where the grand canonical partition function is

$$\mathcal{Z} = \sum_s e^{-\beta E_s - \alpha N_s} \quad (18)$$

- The summation over different states can be done by first considering all the microstates with a given number of particles N_s and then summing up over varying particle number N_s , i.e.

$$\mathcal{Z} = \sum_{N_s=0}^{\infty} e^{-\alpha N_s} \left[\sum_{\text{states with } N_s \text{ particles}} e^{-\beta E_s} \right] \quad (19)$$

Grand canonical partition function

- The inner sum is just the canonical particle function with N_s particles - denote it by $Q_N(\beta, V)$

$$\mathcal{Z}(\beta, \alpha, V) = \sum_{N=0}^{\infty} e^{-\alpha N} Q_N(\beta, V) \quad (20)$$

- The factor $e^{-\alpha}$ is often called as the “fugacity” ($z \equiv e^{-\alpha}$)
- The partition function then is expressed as

$$\mathcal{Z}(\alpha, \beta, V) = \sum_{N=0}^{\infty} z^N Q_N \quad (21)$$

- The average energy becomes

$$\langle E \rangle = U = \sum_s E_s P_s = \frac{-1}{\mathcal{Z}} \frac{\partial \mathcal{Z}}{\partial \beta} \quad (22)$$

Thermodynamics

- Similarly, the average number of particles becomes

$$\bar{N} = \langle N \rangle = -\frac{1}{\mathcal{Z}} \frac{\partial \mathcal{Z}}{\partial \alpha} \quad (23)$$

- Now consider the differential of $q \equiv \ln \mathcal{Z}(\alpha, \beta, V)$

$$dq = \frac{\partial \ln \mathcal{Z}}{\partial \alpha} d\alpha + \frac{\partial \ln \mathcal{Z}}{\partial \beta} d\beta + \frac{\partial \ln \mathcal{Z}}{\partial V} dV \quad (24)$$

$$= -\bar{N}d\alpha - U d\beta + \frac{\partial \ln \mathcal{Z}}{\partial V} dV \quad (25)$$

- All these quantities are exact differentials, and so with some rearrangement they become

$$d[q + \bar{N}\alpha + U\beta] = \alpha d\bar{N} + \beta dU + \frac{\partial \ln \mathcal{Z}}{\partial V} dV \quad (26)$$

$$= \beta \left[dU + \frac{\alpha}{\beta} d\bar{N} + \frac{1}{\beta} \frac{\partial \ln \mathcal{Z}}{\partial V} dV \right] \quad (27)$$

Thermodynamics

- So this becomes

$$k_B T d[q + \bar{N}\alpha + U\beta] = dU + \frac{\alpha}{\beta} d\bar{N} + \frac{1}{\beta} \frac{\partial \ln \mathcal{Z}}{\partial V} dV \quad (28)$$

- Lets look at the last term

$$\frac{1}{\beta} \frac{\partial \ln \mathcal{Z}}{\partial V} = \frac{k_B T}{\mathcal{Z}} \frac{\partial \mathcal{Z}}{\partial V} \quad (29)$$

$$= \frac{k_B T}{\mathcal{Z}} \sum_{N=0}^{\infty} z^N \frac{\partial Q_N}{\partial V} \quad (30)$$

- Recall from canonical ensembles that

$$A = -k_B T \ln Q \quad (31)$$

- and from Maxwell relations that

$$\frac{\partial A}{\partial V} = -P \quad (32)$$

Thermodynamics

- Putting these two together we get

$$\frac{\partial Q_N}{\partial V} = \frac{P}{k_B T} Q_N \quad (33)$$

- Substituting in the last term

$$\frac{1}{\beta} \frac{\partial \ln \mathcal{Z}}{\partial V} = \frac{k_B T}{\mathcal{Z}} \sum_{N=0}^{\infty} \frac{P}{k_B T} z^N Q_N \quad (34)$$

$$= \langle P \rangle \quad (35)$$

- We can denote the average thermodynamic pressure as simply $\langle P \rangle \rightarrow P$
- The expression becomes

$$k_B T d[q + \bar{N}\alpha + U\beta] = dU + PdV + \frac{\alpha}{\beta} d\bar{N} \quad (36)$$

Chemical potential

- Comparing this with the first law of thermodynamics we can identify the entropy as

$$S = k_B[q + \bar{N}\alpha + U\beta] \quad (37)$$

- Now define $\mu \equiv -\alpha/\beta$ to get

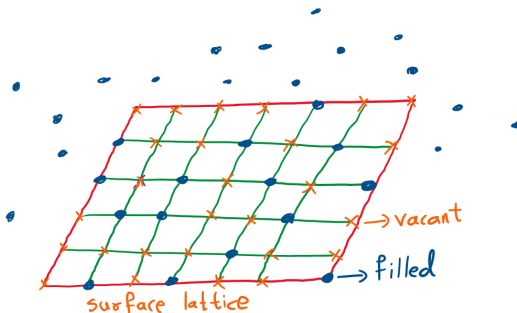
$$TdS = dU + PdV - \mu d\bar{N} \quad (38)$$

- μ is the chemical potential which can be defined as

$$\mu \equiv \left. \frac{\partial U}{\partial \bar{N}} \right|_{S,V} \quad (39)$$

- Its the energy needed to add a particle at constant volume and entropy

Model for adsorption



- Consider N_0 lattice sites on a surface which may be occupied or vacant
- These can correspond to sites where particles can be adsorbed on the surface
- If the site is occupied, its energy is ϵ , or else its energy is 0
- The particles are indistinguishable but the sites are distinguishable
- Exercise: At a given temperature T and chemical potential μ find the fraction of sites occupied

Adsorption

- Calculate the grand canonical partition function

$$\mathcal{Z} = \sum_{N=0}^{N_0} \sum_{\text{states with } N} \exp[-\beta E_s - \alpha N_s] \quad (40)$$

- Given a state where N sites are populated, it has energy $N\epsilon$
- The number of such microstates is $\binom{N_0}{N}$
- Then the grand canonical partition function is

$$\mathcal{Z} = \sum_{N=0}^{N_0} e^{-\alpha N} \binom{N_0}{N} e^{-\beta N\epsilon} \quad (41)$$

$$= \sum_{N=0}^{N_0} \binom{N_0}{N} e^{[-\alpha - \beta\epsilon]N} \quad (42)$$

- This is the binomial expansion of $[1 + e^{[-\alpha - \beta\epsilon]}]^{N_0}$

Adsorption

- The grand canonical partition function becomes

$$\mathcal{Z} = (1 + e^{-\alpha} e^{-\beta\epsilon})^{N_0} = [1 + e^{[\beta(\mu - \epsilon)]}]^{N_0} \quad (43)$$

- To get the average number of particles adsorbed we take

$$\bar{N} = -\frac{1}{\mathcal{Z}} \frac{\partial \mathcal{Z}}{\partial \alpha} \quad (44)$$

$$= -\frac{1}{(1 + e^{-\alpha} e^{-\beta\epsilon})^{N_0}} N_0 (1 + e^{-\alpha} e^{-\beta\epsilon})^{N_0-1} [-e^{-\alpha} e^{-\beta\epsilon}] \quad (45)$$

$$= \frac{N_0}{1 + e^{\alpha} e^{\beta\epsilon}} \quad (46)$$

- Putting $\alpha = -\beta\mu$

$$\bar{N} = \frac{N_0}{1 + e^{[\beta(\epsilon - \mu)]}} \quad (47)$$

Adsorption

- The fraction of adsorption is

$$\frac{\bar{N}}{N_0} = \frac{1}{1 + e^{\beta(\epsilon - \mu)}} \quad (48)$$

- If $\epsilon > \mu$ it will be less than 50% and higher at higher temperatures
- If $\epsilon < \mu$ it will be more than 50% and higher at lower temperatures

Classical ideal gas

- For a classical ideal gas the canonical partition function is of the form

$$Q_1 = \int d\mathbf{p} d\mathbf{q} e^{-\beta \mathbf{p}^2 / 2m} \quad (49)$$

$$= V f(T) \quad (50)$$

- Here $f(T)$ is some function depending on the temperature
- The N -particle canonical partition function is

$$Q_N = \frac{V^N f(T)^N}{N!} \quad (51)$$

- The grand canonical partition function becomes

$$\mathcal{Z} = \sum_{N=0}^{\infty} z^N Q_N = \sum_{N=0}^{\infty} \frac{(zVf)^N}{N!} \quad (52)$$

Classical ideal gas

- This is the Taylor series expansion for the exponential function so it becomes

$$\mathcal{Z} = \exp[zVf] \quad (53)$$

- Now we get

$$P = \frac{1}{\beta} \frac{\partial \ln \mathcal{Z}}{\partial V} = zk_B T f(T) \quad (54)$$

$$\bar{N} = -\frac{1}{\mathcal{Z}} \frac{\partial \mathcal{Z}}{\partial \alpha} = zVf(T) \quad (55)$$

$$U = -\frac{1}{\mathcal{Z}} \frac{\partial \mathcal{Z}}{\partial \beta} = zVk_B T^2 f'(T) \quad (56)$$

- We get back the relation

$$\frac{P}{\bar{N}} = \frac{k_B T}{V} \quad (57)$$

$$(58)$$

Model for solids

- Consider a system of solid material in a grand canonical ensemble
- The single particle partition function would depend on the temperature (the volume is not in phase-space because the particle in a solid is confined at a location)
- Let

$$Q_1 = \phi(T) \quad (59)$$

- Since these particles are distinguishable, so

$$Q_N = \phi(T)^N \quad (60)$$

- The grand partition function becomes

$$\mathcal{Z} = \sum_{N=0}^{\infty} z^N \phi^N \quad (61)$$

Model for solids

- Assuming that $|z\phi| < 1$ we get

$$\mathcal{Z} = \frac{1}{1 - z\phi} \quad (62)$$

- Now we get the thermodynamic properties

$$P = \frac{1}{\beta} \frac{\partial \ln \mathcal{Z}}{\partial V} = 0 \quad (63)$$

- Pressure is not a relevant thermodynamic property for solids

$$\bar{N} = -\frac{\partial \ln \mathcal{Z}}{\partial \alpha} = \frac{z\phi}{1 - z\phi} \quad (64)$$

$$U = -\frac{\partial \ln \mathcal{Z}}{\partial \beta} = \frac{zk_B T^2 \phi'(T)}{1 - z\phi} \quad (65)$$

$$S = k_B [\ln \mathcal{Z} + \bar{N}\alpha + U\beta] = -\bar{N}k_B \ln z - k_B \ln(1 - z\phi) + \frac{zk_B T \phi'}{1 - z\phi} \quad (66)$$

Solid-gas equilibrium

- In order to model solids, $\phi(T)$ can be replaced with the single-particle canonical partition function of classical or quantum simple harmonic oscillators
- Consider a system in a state of equilibrium between solid and gas (for ex. sublimation)
- For the gas phase the fugacity is given by

$$z_g = \frac{\bar{N}_g}{f(T)V_g} \quad (67)$$

- The fugacity for the solid phase is obtained by

$$\bar{N} = \frac{z\phi}{1 - z\phi} \Rightarrow z\phi = \frac{N}{N+1} \approx 1 \quad (68)$$

- N denotes $\bar{N}$ now

Solid-gas equilibrium

- The fugacity of the solid phase will approximately be

$$z_s \approx \frac{1}{\phi(T)} \quad (69)$$

- In equilibrium the fugacity of both phases should match, giving

$$\frac{N_g}{V_g} = \frac{f(T)}{\phi(T)} \quad (70)$$

- If we assume a monotomic ideal gas phase then

$$f(T) = (2\pi mk_B T)^{3/2} / h^3 \quad (71)$$

- If we assume a solid made up of quantum SHMs then

$$\phi(T) = [2 \sinh(\hbar\omega/2k_B T)]^{-3} \quad (72)$$

Solid-gas equilibrium

- Substituting these expressions gives the equilibrium pressure

$$P_g = \frac{N_g}{V_g} k_B T \quad (73)$$

$$= k_B T \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} [2 \sinh(\hbar\omega/2k_B T)]^3 \quad (74)$$

- This gives us an estimate of the equilibrium pressure
- Alternately, if we have a gas of N particles in volume V , this condition gives us a critical temperature when the solid can start forming
- This critical temperature will be given by the implicit relation

$$\frac{f(T_c)}{\phi(T_c)} = \frac{N}{V} \quad (75)$$

- This will be the sublimation/deposition point

Number fluctuations in grand canonical ensemble

- Lets look at the fluctuation in number of particles in a grand canonical ensemble $\Delta N = N - \bar{N}$
- Calculate the rms fluctuation

$$\langle (\Delta N)^2 \rangle = \langle (N - \bar{N})^2 \rangle \quad (76)$$

$$= \langle N^2 \rangle - \bar{N}^2 \quad (77)$$

- Recall that

$$\bar{N} = \frac{\sum_s N_s e^{(-\beta E_s - \alpha N_s)}}{\sum_s e^{(-\beta E_s - \alpha N_s)}} \quad (78)$$

- We can show that

$$\frac{\partial \bar{N}}{\partial \alpha} = \bar{N}^2 - \langle N^2 \rangle \quad (79)$$

Number fluctuations

- From this we get

$$\overline{(\Delta N)^2} = - \frac{\partial \bar{N}}{\partial \alpha} \Big|_{V,T} = k_B T \frac{\partial \bar{N}}{\partial \mu} \Big|_{V,T} \quad (80)$$

- Define the number density n as

$$n \equiv \frac{N}{V}; \quad v \equiv \frac{V}{\bar{N}} = \frac{1}{\bar{n}} \quad (81)$$

- Dividing by V^2 on both sides we get

$$\overline{(\Delta n)^2} = \frac{k_B T}{V} \frac{\partial \bar{n}}{\partial \mu} \Big|_{V,T} = - \frac{k_B T}{v^2 V} \frac{\partial v}{\partial \mu} \Big|_{V,T} \quad (82)$$

Number density fluctuations

- Dividing by $\bar{n}^2$ on both sides gives

$$\frac{(\overline{\Delta n})^2}{\bar{n}^2} = -\frac{k_B T}{V} \frac{\partial v}{\partial \mu} \bigg|_{V,T} = \frac{k_B T}{\bar{N}^2} \frac{\partial \bar{N}}{\partial \mu} \bigg|_{V,T} \quad (83)$$

- Typically the $\bar{N}$ changes slowly with chemical potential μ
- If chemical potential changes by order $k_B T$ then particle number might change by a few, as particles also typically carry $k_B T$ amount of energy
- Therefore, the relative fluctuations in density would be very small
- Exceptions occur near phase transitions, where the small changes in chemical potential can cause particle number to change drastically - this is the regime of large density fluctuations